Grade 8
Unit 11
Lesson 8 Practice

Name Answer Key
Date
Period

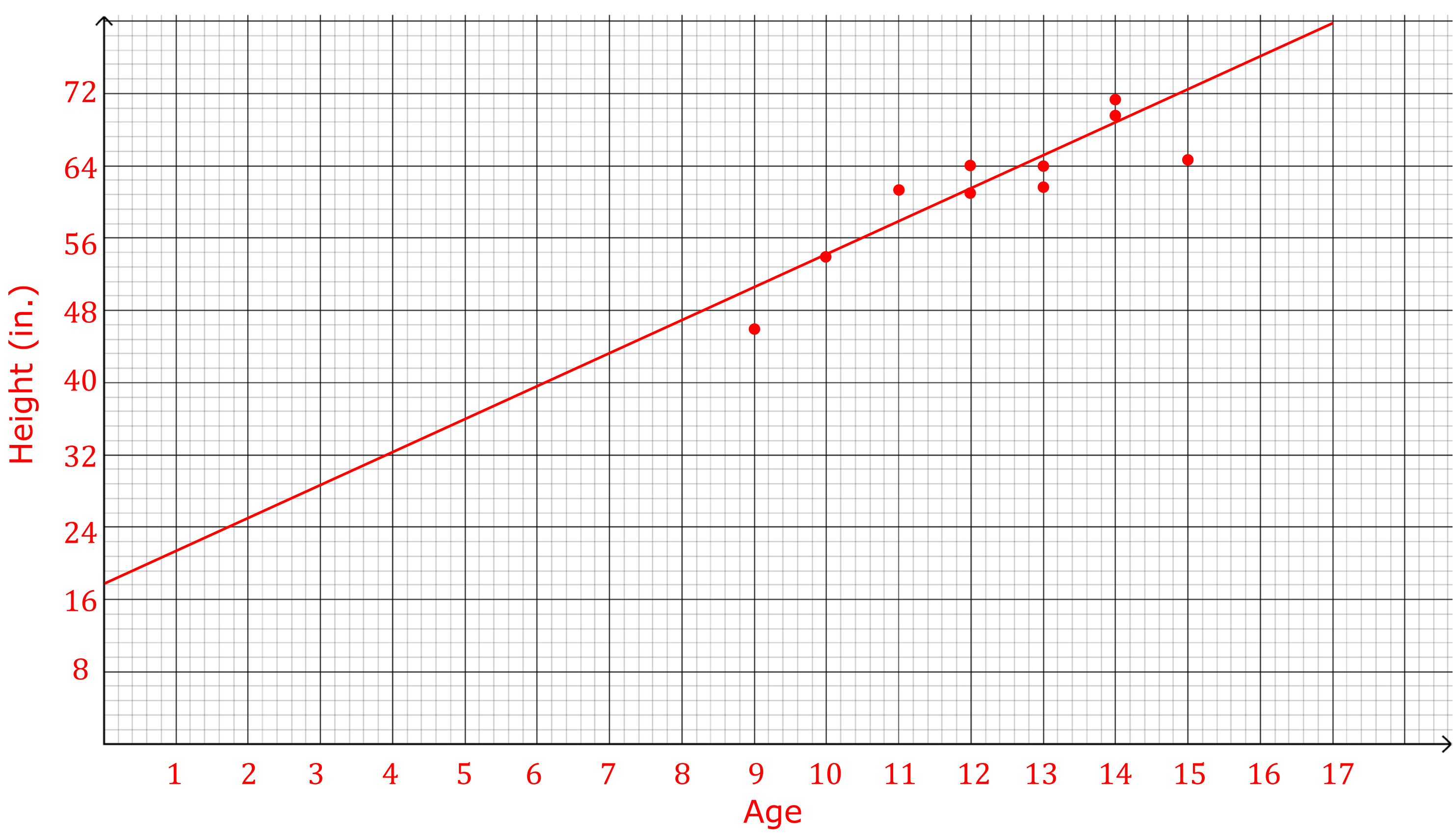
Tomas collected data on a person's age and their height, in inches (in.). His data is shown.

Age	9	10	11	12	12	13	13	14	14	15
Height (in.)	45	53	61	60	64	61	64	69	71	65

Complete the table by determining how best to label the two variables in Tomas' data.

1.	**Dependent Variable**	Height (in.)
2.	**Independent Variable**	Age

A coordinate grid is shown. Relationship between Heights and Ages



3. Determine an appropriate scale that can be used to plot Tomas' data on the coordinate grid. Write that scale on each axis on the coordinate grid.

4. Name the graph and label each axis.

5. Plot the points.

6. Describe any patterns of association in the scatter plot.
Positive, increasing, strong association

7. Informally fit a line to the data using a straightedge.

Tara created a scatter plot for the same data. She fit the line $y = 3.6x + 17.6$ to the data in her scatter plot, where x represents a person's age and y represents the height (in.).

8. Interpret the slope of the line fit to Tara's data.
As the ages increase by 1 year, the height increases by 3.6 in.

9. Explain whether Tara's line would be a good to fit for the data in your scatter plot.
Tara's line has a positive slope as does my line. Also, my line has similar y-intercept to Tara's. Therefore, I think Tara's line would be a good fit.

10. Interpret whether the y-intercept of Tara's equation makes sense in this context.
The intercept can be interpreted that at 0 years, a person is 17.6 in. This could make sense in this context if you think of 0 years as a newborn baby because a height (length) of 17.6 in. is possible.